In [1]:
```
# importing libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
#importing data file
df=pd.read_csv('c:/Users/Lenovo/Documents/mymoviedb.csv', lineterminator='\n')
df.head()
```

Out[2]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language	Genre	
0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en	Action, Adventure, Science Fiction	https://image.tmdb.o
1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en	Crime, Mystery, Thriller	https://image.tmdb.org
2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en	Thriller	https://image.tmdb.org/
3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en	Animation, Comedy, Family, Fantasy	https://image.tmdb.org
4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en	Action, Adventure, Thriller, War	https://image.tmdb.org

In [13]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Release_Date       9827 non-null   object 
 1   Title              9827 non-null   object 
 2   Overview           9827 non-null   object 
 3   Popularity         9827 non-null   float64
 4   Vote_Count         9827 non-null   int64  
 5   Vote_Average       9827 non-null   float64
 6   Original_Language  9827 non-null   object 
 7   Genre              9827 non-null   object 
 8   Poster_Url         9827 non-null   object 
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB
```

In [15]:
```
#exploring stats
df.describe()
```

Out[15]:

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805536	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

In [17]:
```
#exploring genres
df['Genre'].head()
```

Out[17]:
```
0    Action, Adventure, Science Fiction
1             Crime, Mystery, Thriller
2                             Thriller
3    Animation, Comedy, Family, Fantasy
4      Action, Adventure, Thriller, War
Name: Genre, dtype: object
```

In [19]:
```
#checking duplicates
df.duplicated().sum()
```

Out[19]:
```
0
```

Data cleaning

In [7]:
```
# casting release date column

df['Release_Date']=pd.to_datetime(df['Release_Date'])


print(df['Release_Date'].dtypes)
```

```
datetime64[ns]
```

In [9]:
```
df['Release_Date']=df['Release_Date'].dt.year
df['Release_Date'].dtypes
```

Out[9]:
```
dtype('int32')
```

In [36]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Release_Date       9827 non-null   int32
 1   Title              9827 non-null   object
 2   Overview           9827 non-null   object
 3   Popularity         9827 non-null   float64
 4   Vote_Count         9827 non-null   int64
 5   Vote_Average       9827 non-null   float64
 6   Original_Language  9827 non-null   object
 7   Genre              9827 non-null   object
 8   Poster_Url         9827 non-null   object
dtypes: float64(2), int32(1), int64(1), object(5)
memory usage: 652.7+ KB
```

In [11]:
```
df.head()
```

Out[11]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language	Genre	
0	2021	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en	Action, Adventure, Science Fiction	https://image.tmdb.o
1	2022	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en	Crime, Mystery, Thriller	https://image.tmdb.org
2	2022	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en	Thriller	https://image.tmdb.org/
3	2021	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en	Animation, Comedy, Family, Fantasy	https://image.tmdb.org
4	2021	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en	Action, Adventure, Thriller, War	https://image.tmdb.org

Dropping unwanted columns

In [15]:
```
# list of columns that need to be dropped
```

```
cols=['Overview','Original_Language','Poster_Url']
```

In [17]:
```
#dropping columns
df.drop(cols,axis=1,inplace=True)
df.columns
```

Out[17]:
```
Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count', 'Vote_Average',
       'Genre'],
      dtype='object')
```

In [ ]:
```
edges = [1, 3.25, 5.5, 7.75, 10]
labels = ['not_popular', 'below_avg', 'average', 'popular']
df['Vote_Average'] = pd.cut(df['Vote_Average'], bins=edges, labels=labels, include_lowest=True)
```

In [61]:
```
#confirming changes
df.head()
```

Out[61]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	below_avg	Thriller
3	2021	Encanto	2402.201	5076	popular	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	average	Action, Adventure, Thriller, War

In [64]:
```
df['Vote_Average'].unique()
```

Out[64]:
```
['popular', 'below_avg', 'average', 'not_popular', NaN]
Categories (4, object): ['not_popular' < 'below_avg' < 'average' < 'popular']
```

In [66]:
```
# exploring column
df['Vote_Average'].value_counts()
```

Out[66]:
```
Vote_Average
not_popular    2467
popular        2450
average        2412
below_avg      2398
Name: count, dtype: int64
```

In [70]:
```
# dropping Nan's
df.dropna(inplace=True)

#confirming changes
df.isna().sum()
```

Out[70]:
```
Release_Date    0
Title           0
Popularity      0
Vote_Count      0
Vote_Average    0
Genre           0
dtype: int64
```

In [74]:
```
df['Genre'].describe()
```

Out[74]:
```
count      9727
unique     2324
top       Drama
freq        458
Name: Genre, dtype: object
```

splitting Genre column and exploding dataframe one genre per row as it will be easy to access

In [79]:
```
df['Genre']= df['Genre'].str.split(', ')

#exploding the data frame
df=df.explode('Genre').reset_index(drop=True)
df.head()
```

Out[79]:

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction
3	2022	The Batman	3827.658	1151	popular	Crime
4	2022	The Batman	3827.658	1151	popular	Mystery

In [83]:
```
#casting column into category datatype
df['Genre']=df['Genre'].astype('category')

#confirming changes

df['Genre'].dtypes
```

Out[83]:
```
CategoricalDtype(categories=['Action', 'Adventure', 'Animation', 'Comedy', 'Crime',
                  'Documentary', 'Drama', 'Family', 'Fantasy', 'History',
                  'Horror', 'Music', 'Mystery', 'Romance', 'Science Fiction',
                  'TV Movie', 'Thriller', 'War', 'Western'],
 , ordered=False, categories_dtype=object)
```

In [85]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Release_Date  25552 non-null  int32
 1   Title         25552 non-null  object
 2   Popularity    25552 non-null  float64
 3   Vote_Count    25552 non-null  int64
 4   Vote_Average  25552 non-null  category
 5   Genre         25552 non-null  category
dtypes: category(2), float64(1), int32(1), int64(1), object(1)
memory usage: 749.6+ KB
```

In [93]:
```
df.nunique()
```

Out[93]:
```
Release_Date     100
Title           9415
Popularity      8088
Vote_Count      3265
Vote_Average       4
Genre             19
dtype: int64
```

Data visualization

In [96]:
```
#setting seaborn configurations

sns.set_style('dark')
```

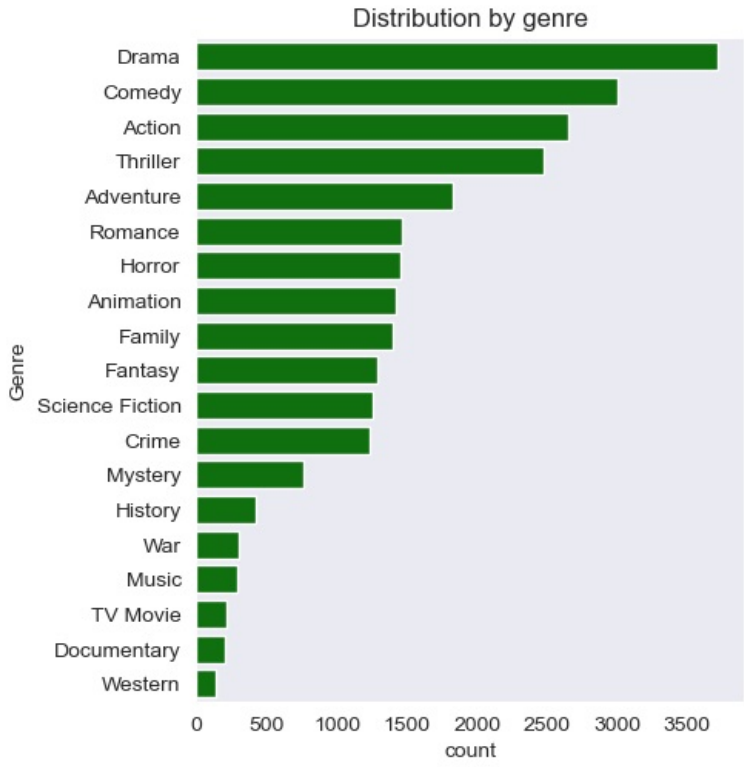
What is the most frequent genre of movies released on Netflix?

In [103…
```
df['Genre'].describe()
```

Out[103…
```
count     25552
unique       19
top       Drama
freq       3715
Name: Genre, dtype: object
```

In [113…
```
#visual presentation

sns.catplot(data=df,kind='count',y='Genre',order=df['Genre'].value_counts().index,color='g')
plt.title('Distribution by genre')
plt.show()
```



Which movie got the highest popularity ? what's its genre ?

In [165…

```
# checking the most popular movie and its genre
df[df['Popularity'] == df['Popularity'].max()]
```

Out[165…

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction

What movie got the lowest popularity? what's its genre?

In [168…

```
#checking the least popular movie and its genre
df[df['Popularity'] == df['Popularity'].min()]
```

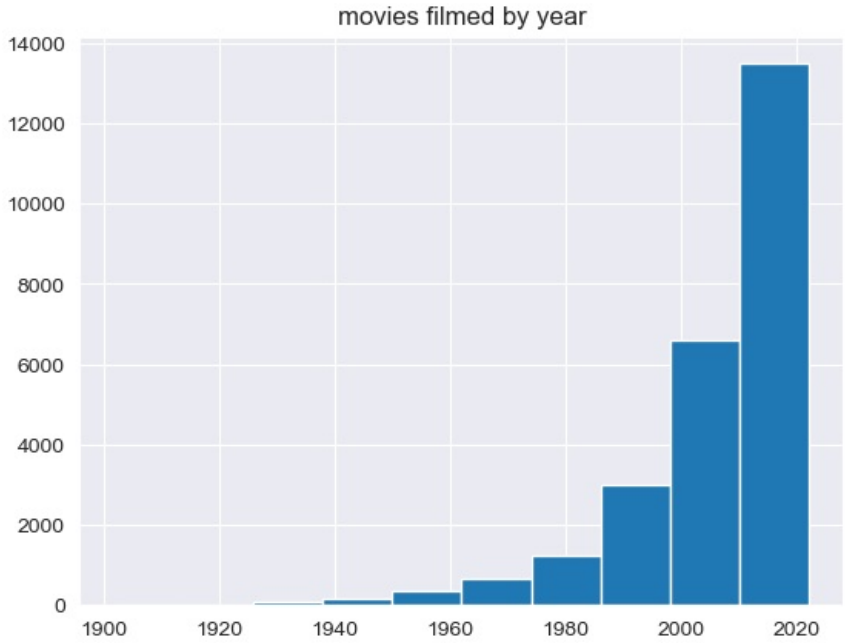
Out[168…

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
25546	2021	The United States vs. Billie Holiday	13.354	152	average	Music
25547	2021	The United States vs. Billie Holiday	13.354	152	average	Drama
25548	2021	The United States vs. Billie Holiday	13.354	152	average	History
25549	1984	Threads	13.354	186	popular	War
25550	1984	Threads	13.354	186	popular	Drama
25551	1984	Threads	13.354	186	popular	Science Fiction

Q5: Which year has the most filmmed movies?

In [171…

```
df['Release_Date'].hist()
plt.title('movies filmed by year')
plt.show()
```



In [ ]:

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